1. What is CSS and why is it used?

Expected Answer: CSS (Cascading Style Sheets) is used to style and layout HTML elements. It controls visual aspects like colors, fonts, spacing, and responsiveness, separating content from presentation.

2. What are the different types of CSS?

Expected Answer:

- Inline CSS: written inside the HTML element
- Internal CSS: written in a <style> tag within the HTML file
- External CSS: written in a separate .css file and linked via <link>

3. What is the difference between class and id selectors in CSS?

Expected Answer: id is unique and targets a single element (#id), while class can be reused across multiple elements (.class). id has higher specificity.

4. What is the Box Model in CSS?

Expected Answer: The Box Model consists of:

- Content
- Padding
- Border
- Margin It defines how elements are spaced and sized on the page.

5. What is specificity in CSS?

Expected Answer: Specificity determines which CSS rule is applied when multiple rules target the same element. Inline styles > IDs > Classes > Elements.

6. What is the difference between relative, absolute, and fixed positioning?

Expected Answer:

- relative: positioned relative to its normal position
- absolute: positioned relative to the nearest positioned ancestor
- fixed: positioned relative to the viewport and doesn't scroll

7. What is the z-index property used for?

Expected Answer: z-index controls the stacking order of elements. Higher values appear on top of lower ones when elements overlap.

8. What are pseudo-classes in CSS? Give examples.

Expected Answer: Pseudo-classes define a special state of an element. Examples:

- :hover (when mouse is over)
- :focus (when input is focused)
- :nth-child(n) (targets specific children)

9. What is the difference between em, rem, and px units?

Expected Answer:

- px: absolute unit
- em: relative to parent's font size
- rem: relative to root (html) font size

10. What is Flexbox in CSS3?

Expected Answer: Flexbox is a layout model that allows responsive alignment and distribution of space among items in a container. It simplifies vertical and horizontal centering.

11. What are media queries in CSS3?

Expected Answer: Media queries allow CSS to adapt styles based on device characteristics like screen size, resolution, or orientation. Example: @media (max-width: 768px) { … }

12. What is the difference between visibility: hidden and display: none?

Expected Answer:

- visibility: hidden: hides the element but keeps its space
- display: none: removes the element and its space from layout

13. What is the calc() function in CSS3?

Expected Answer: calc() allows mathematical calculations in CSS. Example: width: calc(100% - 50px); combines percentages and pixels dynamically.

14. What is the difference between min-width, max-width, and width?

Expected Answer:

- width: sets exact width
- min-width: sets minimum width
- max-width: sets maximum width Used for responsive design control.

15. What is the transition property in CSS3?

Expected Answer: transition allows smooth changes between property values. Example: transition: all 0.3s ease; animates changes in style over time.

16. What is the transform property in CSS3?

Expected Answer: transform applies 2D or 3D transformations like rotate, scale, skew, or translate. Example: transform: rotate(45deg);

17. What is the opacity property in CSS?

Expected Answer: opacity controls transparency. Value ranges from 0 (fully transparent) to 1 (fully opaque).

18. How do you center an element horizontally and vertically using Flexbox?

Expected Answer:

```
display: flex;
justify-content: center;
align-items: center;
```

19. What is the difference between inline, block, and inline-block elements?

Expected Answer:

- inline: flows with text, no width/height
- block: takes full width, starts on new line
- inline-block: behaves like inline but allows width/height

20. What is the grid layout in CSS3?

Expected Answer: CSS Grid is a two-dimensional layout system. It allows precise control over rows and columns. Example: display: grid; grid-template-columns: 1fr 2fr;

21. What is the object-fit property used for?

Expected Answer: object-fit defines how media (like images or videos) should be resized to fit their container. Example: object-fit: cover;

22. What is the clamp() function in CSS?

Expected Answer: clamp() sets a value between a minimum, preferred, and maximum range. Example: font-size: clamp(1rem, 2vw, 2rem);

23. What is the difference between inherit, initial, and unset in CSS?

Expected Answer:

- inherit: inherits value from parent
- initial: resets to default value
- unset: removes applied value (acts as inherit or initial)

24. What is the :root selector in CSS?

Expected Answer: :root targets the highest-level element (<html>). Commonly used to define global CSS variables.

25. How do you define and use CSS variables?

Expected Answer:

```
:root {
  --main-color: #3498db;
}
element {
  color: var(--main-color);
}
```

26. What is the overflow property in CSS?

Expected Answer: overflow controls content that exceeds container boundaries. Values: visible, hidden, scroll, auto.

27. What is the difference between absolute and relative units in CSS?

Expected Answer: Absolute units (px, cm) are fixed. Relative units (em, %, vw) scale based on parent or viewport, aiding responsiveness.

28. What is the animation property in CSS3?

Expected Answer: animation allows keyframe-based animations.

29. What is the difference between visibility: hidden and opacity: 0?

Expected Answer: Both hide the element visually, but visibility: hidden prevents interaction, while opacity: 0 keeps it interactive unless pointer events are disabled.

30. How do you make a responsive layout using CSS3?

Expected Answer: Use media queries, Flexbox, Grid, relative units (%, vw, em), and avoid fixed widths. Combine with mobile-first design principles.